EL NIÑO ACTION BRIEFING · 30 OCTOBER 2026

Nai Wiang, Mueang Nan, Nan

Smoke and heat peak together in March: give every bedridden or homebound heart and lung patient one filtered room that is not left to overheat, before February.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Nan province (tracker, province level)	43 forecast for Nan, 29 of them in March; 14 already past (Jun–Sep)	6-month forecast, Oct 2026 – Mar 2027	Dry-season smoke (PM2.5) and heat, February–April; river flood in the wet season

11/11 recommendations checked. 5 corrected, 1 unverified, 5 contraindicated (the five "Do not do this" cards; one of them, a pre-signed disaster trigger, was found in verification). All 9 For you steps were checked too, and 7 were corrected. Exposure figure: Climate Impact Lab El Niño Heat Mortality Tracker data, version of 26 October 2026, pre-peer-review, methods forthcoming. It is published for Nan province, not for Nai Wiang, and it counts heat only. Smoke is this place's other large risk, and no comparable figure exists for it. At 43 forecast deaths in a province of about 476,000, the toll will not be visible in local statistics; judge these actions by whether the rooms and visits happen, not by a body count. The funding target fell from the panel's THB 5.5 million (US$163,900) to THB 1.8 million (US$53,500) after verification (−67%), because the Department of Health is already building clean rooms in Nan and the Ministry of Public Health already hands out masks.

EXECUTIVE SUMMARY — AT A GLANCE



18.78°N 100.78°E, Thailand · Natural Earth

- **The risk.** Nan sits in a mountain bowl that traps smoke; in March 2025 parts of the province were locked down, and in 2026 burning was banned for 47 days. The March heat arrives in the same weeks.
- **For you.** As a resident, by 15 January build a fan-and-filter purifier (about THB 1,200–1,500) for the bedroom of the oldest or sickest person in the house.
- **The money.** Floor THB 0.48 million (US$14,100), target THB 1.8 million (US$53,500), ceiling THB 5.4 million (US$160,300); filtered rooms for bedridden heart and lung patients come first.
- **Do not.** Do not shut an older person into a sealed, un-cooled room on a March or April afternoon to escape the smoke; the heat builds up.*

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* Suggested formulation: check with a health worker before sharing.

FOR YOU, AS A RESIDENT 9 STEPS

In Nai Wiang the danger is a March afternoon when the smoke is thick and the house is hot, for an older person with a weak heart or lungs who lives in an open wooden house. The ranked actions below are written for the district and its partners. These are the ones you can take yourself.

Build a fan-and-filter purifier for the bedroom of the oldest or sickest person in the house

PARTIALLY CONFIRMED FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
15 January 2027	About THB 1,200–1,500 (US$36–45)	Anyone with heart or lung disease, small children

Portable filtration cut personal PM2.5 by 53% and lowered systolic blood pressure by 3.2 mmHg in 40 older adults in Detroit (Morishita et al. 2018). Suan Dok Hospital in Chiang Mai built more than 500 do-it-yourself units at THB 1,200–1,500 each, its own production cost (Chiang Mai CityLife). Run it with the windows shut on red air-quality days.

Buy fitted N95 or KN95 masks for each adult, and learn to fit them

PARTIALLY CONFIRMED DO-NOT 3

BY WHEN	COST TO YOU	PROTECTS
31 January 2027	About THB 350–460 (US$10–14) for 20 masks	Anyone who must go outdoors on smoke days

The Department of Health advises N95 or KN95 masks worn correctly (Nation Thailand); cloth masks filter about half as well. Thai retail prices run about THB 18–23 a mask (ProductNation). Ask your health centre first, because the Ministry of Public Health distributes masks to at-risk groups.

Collect refills of inhalers and heart medicines before the smoke season *

PARTIALLY CONFIRMED FROM ACTION 3

BY WHEN	COST TO YOU	PROTECTS

* Suggested formulation: check with a health worker before sharing.

31 January 2027	Usually nothing under Universal Coverage	People with asthma, COPD, or heart disease

Admissions for lung disease rise in the days after a smoke peak, so the medicine has to be in the house before it. Under the universal coverage scheme, people aged 60 and over were exempt from the co-payment, and the co-payment was largely dropped after 2006–07; we could not confirm the current rule on the NHSO site, so ask your health centre.

Do not shut yourself into a sealed room on a hot afternoon: open it and cool down when the room gets hot *

PARTIALLY CONFIRMED · DO-NOT 1

BY WHEN	COST TO YOU	PROTECTS
March–April 2027	None	Older people indoors

A sealed room without air conditioning keeps the smoke out and the heat in. The Department of Health's heat advice is to stay somewhere shaded and well ventilated. In March and April, run the purifier in the morning and evening, and in the hottest hours wet the skin and move to a cooler place if the room heats up.*

Do field and outdoor work before 11:00, drink water rather than alcohol, and rest in the shade from 11:00 to 15:00 *

CONFIRMED · ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
March–May 2027	None	Farmers, labourers, vendors

Heatstroke killed 61 people in Thailand from January to April 2024; 62% of them drank alcohol regularly, and most were farmers or labourers. The Department of Disease Control advises avoiding outdoor activity from 11:00 to 15:00 and avoiding alcohol (Nation Thailand).*

Install the Air4Thai app and act on the Nan Municipality station

CONFIRMED · FROM ACTION 2

BY WHEN	COST TO YOU	PROTECTS

* Suggested formulation: check with a health worker before sharing.

31 January 2027	None	The whole household

Air4Thai reports PM2.5 hourly and as a 24-hour average, and Nan has two stations, at Nan Municipality and at Chaloem Phra Kiat Hospital (Thansettakij). Emergency number: 1669. Department of Health hotline: 1478. Department of Disease Control: 1422.

Give your village health volunteer the name of any bedridden or elderly neighbour who lives alone

PARTIALLY CONFIRMED · FROM ACTION 2

BY WHEN	COST TO YOU	PROTECTS
15 December 2026	None	Older people living alone

Village health volunteers already have a national role in checking six heat-risk groups (Thansettakij). A name on their list is what brings someone to the door on a red day. Being told to stay indoors protects no one who lives alone in a house with gaps in the floorboards.

Empty standing water around the house now, and again in April and May

PARTIALLY CONFIRMED · ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
30 November 2026, then April–May 2027	None	The household, against dengue

Nan recorded 2,881 dengue cases in 2023, 605.8 per 100,000 people (Nan Provincial Health Office). The August 2026 flood left debris that holds water. The main dengue season is May to September, so repeat the clean-up before it.

Check that your state welfare card is still active, now that the extra THB 700 has ended

PARTIALLY CONFIRMED · ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
31 December 2026	None	Household food budget

* Suggested formulation: check with a health worker before sharing.

The card's base credit is THB 300 a month; the THB 700 top-up ran from 1 June to 30 September 2026 and has ended (Thansettakij). The 2026 registration results were issued on 17 July, with appeals in September; if you were left off, ask the district office now.

What to ask of the people in charge

- Ask the Nan Town Municipality's Local Health Security Fund committee to approve filtered rooms for the bedridden and homebound heart and lung patients in your community before the end of December (action 1).
- Ask the Nan Provincial Health Office how many clean rooms Nan has already received from the Department of Health, and whether any are in homes (action 1).
- Ask the Governor's office to publish a standing haze operating plan by January, with its forest-fire schedule, rather than an emergency order after the province turns red (action 5).

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet.

1. Put a filtered room in the home of every bedridden or homebound heart and lung patient in Nai Wiang by 31 January 2027, funded through the municipality's Local Health Security Fund

PARTIALLY CONFIRMED · PUBLIC HEALTH · PHILANTHROPIST · EMERGENCY MANAGER · HISTORIAN · FRONTLINE · SKEPTIC · ADAPTATION INVENTORY: NOT QUERYABLE

OWNER	BY WHEN	COST	PREVENTS
Nan Town Municipality, through its Local Health Security Fund, with Nan Hospital	31 January 2027	THB 1.31 million (US$38,800) for about 300 homes, estimate	Heart and lung deaths among people who cannot leave home on smoke days

* Suggested formulation: check with a health worker before sharing.

The National Health Security Office confirmed on 30 March 2026 that local funds can buy PM2.5 masks and pay for dust-free areas for children, older people, and the chronically ill (MGR Online). Chiang Mai University's dust-free room costs THB 3,600 and combines positive-pressure fresh air, a do-it-yourself purifier, and a sensor (Thairath); Suan Dok Hospital's purifier alone costs THB 1,200–1,500. Filtration cut personal PM2.5 by 53% in older adults (Morishita 2018). In Chiang Mai each 10 μg/m³ of PM2.5 raised non-accidental deaths by about 1.6% (PMC6687987).

Corrected in verification: the university's rooms have so far gone to 83 care institutions, not homes, so this extends the design; the claim that Nan is a pilot province has no source. The evidence measures exposure and blood pressure, not deaths. The 300 homes is an estimate; the real count is in the provincial long-term-care register. The Department of Health's 2026 programme reached 2,275 rooms in 10 high-risk provinces including Nan, so check what Nan already has before buying.

2. Have village health volunteers visit each room-holder and each older person living alone weekly from February to April, and daily on red-AQI or 40°C days

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH · PHILANTHROPIST · FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT QUERYABLE

OWNER	BY WHEN	COST	PREVENTS
Nan Provincial Health Office, through village health volunteers	List by 15 December 2026; visits February–April 2027	THB 195,000 (US$5,800) for visit stipends, estimate	Deaths of people alone at home, and missed meals on days they cannot reach the market

Village health volunteers already hold a national role in heat surveillance of six risk groups (Thansettakij), and screened residents in past Nan haze episodes. A visit checks the purifier, the person's temperature, medicines, and food, and refers to Nan Hospital if needed. The municipality's lists of households flooded in 2006 and 2024 are a starting register for riverside homes.

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: a weekly visit routine for room-holders is a new local design. Heat-warning evidence is weaker than the panel claimed: Philadelphia's 4.4% fall in deaths was the only benefit among 20 US cities, and the Hong Kong figure could not be confirmed. The panel's separate food-delivery roster is merged here.

3. Run a January sweep at Nan Hospital and the health centres: refill inhalers and heart medicines and give each heart and lung patient a written plan for smoke days

UNVERIFIED · PUBLIC HEALTH · ADAPTATION INVENTORY: NOT QUERYABLE

OWNER	BY WHEN	COST	PREVENTS
Nan Hospital	31 January 2027	Within existing universal coverage budgets	Emergency admissions in the week after a smoke peak

Doctors at Nan Hospital named heart and lung patients as most at risk after more than 8,000 smoke-related visits in under a month during an earlier haze episode (Bangkok Post). Chiang Mai studies report emergency visits for lung disease lagging the smoke peak.

Unverified: the lag study and Nan Hospital's capacity were not checked to a primary source. It is a sound clinical precaution with no local evaluation behind it.

4. Prepare a filtered ward and a heatstroke cooling protocol at Nan Hospital for March–April 2027

PARTIALLY CONFIRMED · HISTORIAN · EMERGENCY MANAGER · ADAPTATION INVENTORY: NOT QUERYABLE

OWNER	BY WHEN	COST	PREVENTS
Nan Hospital	28 February 2027	THB 300,000 (US$8,900), estimate, within the hospital budget	Deaths among the patients who arrive in March

The hospital has had to move patients in floods twice in two years: in July 2025, with water up to 1.83 m, ventilator and dialysis patients were moved (Thaiger), and in August 2026 it ran a field hospital. The money and attention since 2024 have gone to water. The tracker puts 29 of the province's 43 forecast heat deaths in March, the month the smoke peaks.

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: the cost is the panel's estimate; no hospital plan for heat or smoke was found to confirm or rule out. Payer: Nan Hospital, through its own budget.

5. Agree a standing haze operating plan by January: forest-fire schedule and firebreaks before the ban, published monthly hotspot counts, and a pre-agreed procedure for the disaster declaration

PARTIALLY CONFIRMED | EMERGENCY MANAGER · ECONOMIST · SKEPTIC · HISTORIAN | ADAPTATION INVENTORY: NOT QUERYABLE

OWNER	BY WHEN	COST	PREVENTS
Nan Governor, with the provincial DDPM and forest units	31 January 2027	Administrative; firebreak crews unfunded	Another year of orders issued after the province is already red

In 2026 Nan banned burning from 15 March to 30 April in all 15 districts (Daily News), and closed nine forest reserves only on 4 April (Nation Thailand). The Ministry of Finance raised the governor's emergency advance to THB 50 million per province from November 2025, released after a disaster declaration. A haze declaration requires PM2.5 of 150 µg/m³ or more for five consecutive days (Thai PBS); Chiang Mai, Lamphun, and Phayao have used it. A written procedure lets Nan declare on the day the data qualifies.

Corrected in verification: a governor cannot pre-sign a declaration, and the panel's 75 µg/m³ threshold is Bangkok's nuisance rule, not the disaster criterion (see do-not 2). Village fire crews worked here before: in 1998, 45 Nan villages cut 94 km of firebreaks with 765 fireguards (RECOFTC); RECOFTC's current work in four Doi Phu Kha villages has no confirmed funding after 2024.

6. Set a dry-season water trigger with the Provincial Waterworks Authority: if the Nan River intake falls below a stated level, water trucks go to the riverside communities

PARTIALLY CONFIRMED | EMERGENCY MANAGER | ADAPTATION INVENTORY: NOT QUERYABLE

OWNER	BY WHEN	COST	PREVENTS
Provincial Waterworks Authority, Nan branch	31 January 2027	Contingency only; no cost until the trigger is met	A repeat of the March 2012 drought shortage

* Suggested formulation: check with a health worker before sharing.

In March 2012 all 15 Nan districts were declared drought areas, with about 216,000 people short of water and about THB 59 million in losses (Pattaya Mail). After the August 2026 flood clogged the intake, the Waterworks Authority kept supplying at low pressure and restored full supply within days (PRD).

Corrected in verification: town water comes from the Provincial Waterworks Authority, not the municipality, and supply was not left unrestored after the flood, so the panel's THB 500,000 truck contract has no current evidence base. Keep it only as a contingency tied to a measured level.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Nan Municipality LHSF	Nan Provincial Health Office	Nan Hospital	Village health volunteers	Nan Governor	Provincial DDPM	Forest and fire units	PWA Nan
1. Filtered rooms for homebound heart and lung patients	**A**	**C**	**R**	**R**				
2. Volunteer visits on smoke and heat days	**C**	**A**	**I**	**R**				
3. January medicine sweep		**R**	**A**	**I**				
4. Hospital ready for heat and smoke patients		**C**	**A**					
5. A standing haze plan, not an emergency order		**C**			**A**	**R**	**R**	
6. Water: a trigger, not a contract					**I**	**R**		**A**

What the chart shows: The health system carries four of six rows, and the municipality's health fund is the one place money can land quickly. Two rows (the haze plan and the water trigger) need a decision, not money. The village health volunteers do the work in three rows and are accountable for none.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

THB 0.48 million (US$14,100)

TARGET

THB 1.8 million (US$53,500)

CEILING

THB 5.4 million (US$160,300)

LARGEST UNCERTAINTY

How many bedridden and homebound patients Nai Wiang has, and how many clean rooms Nan already received

Floor = the 100 most severe homes, 100 × (THB 3,600 room + 500 spare filter) = 410,000, plus their visits, 100 × 13 weeks × THB 50 = 65,000, so THB 475,000. Target = about 300 homes, 300 × 4,100 = 1,230,000 plus a filter reserve of 77,000, + visits for all 300, 195,000, + the hospital ward, 300,000, so THB 1,802,000. Ceiling = target + 1,000 more homes × THB 3,600 = 3.6 million, so THB 5.4 million; one hospital built about 500 units in a season, so more than this cannot be installed before February. Every unit cost is an estimate. The panel's target was THB 5.5 million; it fell by 67%, because the extra 1,000 rooms duplicate the Department of Health's programme until Nan's allocation is known, the masks duplicate Ministry of Public Health stock, and the fire-crew line's condition (a lapse in RECOFTC's funding) was not met.

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Nan Town Municipality Local Health Security Fund, with Nan Hospital	Filtered rooms and spare filters for about 300 homebound heart and lung patients (action 1)	THB 1.31 million (US$38,800)	Matching grant into the fund	Estimate; PM2.5 −53% and blood pressure −3.2 mmHg in older adults (Morishita 2018); no cost-per-death study	PARTIALLY CONFIRMED
Nan Provincial Health Office (village health volunteer stipends)	Weekly visits February–April (action 2)	THB 195,000 (US$5,800)	Top-up to an existing scheme	Estimate, 300 × 13 × THB 50; a room only protects someone who uses it	PARTIALLY CONFIRMED
Nan Hospital	Filtered ward and cooling protocol (action 4)	THB 300,000 (US$8,900)	Direct grant	Estimate; no published unit cost	PARTIALLY CONFIRMED
Nan Town Municipality Local Health Security Fund	A further 1,000 homes, only after Nan's Department of Health allocation is known (action 1)	THB 3.6 million (US$106,800)	Matching grant	Estimate; possible duplicate of the Department of Health programme	UNVERIFIED

Money through the municipality's health fund is slower than a direct grant, but it leaves an approved budget line the municipality can run again next year. The returns flatten once every bedridden and homebound patient has a room; after that, money buys masks for people who were already coping.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

Do not shut an older person into a sealed, un-cooled room on a hot afternoon to escape the smoke *

CONTRAINDICATED

March brings the worst smoke and the start of the heat together. A sealed room without air conditioning keeps the smoke out and the heat in, and the Department of Health's heat advice is to stay somewhere shaded and ventilated. In a trial of adults aged 65–85 at 36°C, an electric fan did not lower core temperature at all (JAMA 2025).

Instead: Run the purifier in the cooler hours, use a design that brings in filtered fresh air, and in the hottest hours wet the skin and move to a cooler place.

Do not pre-sign a disaster declaration on a 75 µg/m³ trigger

CONTRAINDICATED

The panel proposed this; verification contraindicated it. A governor cannot commit to a declaration in advance, because each one must rest on the facts of the day, and 75 µg/m³ for three days is Bangkok's nuisance-control rule, not the disaster criterion of 150 µg/m³ for five days. An order on the wrong threshold could be challenged, and the money would stall.

Instead: Write the standard operating procedure now (action 5), so the declaration can be made the day the data qualifies.

Do not hand out cloth or surgical masks for smoke

CONTRAINDICATED

In fit tests, homemade cotton masks filtered a median of 50%, against 80% for surgical masks, and fine smoke particles leak around a loose mask. The Department of Health advises fitted N95 or KN95 masks.

* Suggested formulation: check with a health worker before sharing.

Instead: Give fitted N95 or KN95 masks to outdoor workers, carers, and heart and lung patients, and show them how to fit one.

Do not spend the haze budget on water-spraying trucks or cloud seeding

CONTRAINDICATED

Chulalongkorn engineers class water spraying as ineffective. In March 2025 Chiang Mai sprayed about 16,000 litres four times a day while PM2.5 stayed above the standard in 25 districts, and no evaluation shows that cloud seeding cuts PM2.5.

Instead: Put that money into filtered rooms for the people most at risk (action 1).

Do not lengthen the blanket burning ban as the main fix

CONTRAINDICATED

In Chiang Mai people burned before and after the zero-burn window, which lengthened the smoke season, and Chiang Mai moved to permits. Much of the smoke over Nan blows in from northern Laos, which a Nan ban cannot reach, and enforcement lands on hill smallholders while maize contracts drove the clearing.

Instead: Schedule managed burning and firebreaks before the ban, publish hotspot counts by month so any shift shows, and fund alternatives to maize burning.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2006	May flash floods and landslides across five northern provinces, mostly in Uttaradit	48–87 dead across five provinces; no Nan-only toll found	Flood walls along the Nan River

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2012	Drought: all 15 Nan districts declared disaster areas (27–28 March)	About 216,000 people short of water; about THB 59 million in losses	Not found
2023	Dengue outbreak	2,881 cases, 605.8 per 100,000; no deaths	Not found
2024	After the 2023–24 El Niño: heatstroke kills 61 in Thailand from January to April. In August the Nan River crests at 8.72 m, 30 cm above 2006	61 heatstroke deaths nationally; 3 flood deaths in Nan	Flood lists kept by the municipality
2025	March smoke: AQI 152; lockdown in three districts and a curfew in Sanian, Mueang Nan. July: storm Wipha's remnants flood the town; Nan Hospital moves patients	Not reported	Emergency orders, not a standing plan
2026	April smoke: nine forest reserves closed on 4 April; PM2.5 of 210.9 μg/m³, the highest of ten risk provinces. August flood	3 flood deaths; 67,129 people affected	Burning banned 15 March – 30 April

The recurrence nobody has noticed: the bare maize hillsides that send water into Nan town each August are the same fields that burn each March. Since 2024 the money and attention have gone to floods, and each spring has been met with a new emergency order rather than a standing plan. This El Niño turns the wettest years into the driest one.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Issue the burning ban and forest closures earlier, from 1 January** (emergency manager) versus **bans only move the burning to other dates** (skeptic). The 2026 orders came after the province turned red, which supports earlier action, but Chiang Mai's experience is that bans lengthen the season. Monthly hotspot counts before, during, and after the ban would settle it here (action 5).
- **A purifier in every home** (public health scientist, philanthropist) versus **a sealed room is a heat trap in March** (frontline, verification). Both are right; the answer is a filtered room with fresh air, used in the cooler hours, with a visit on hot days.

* Suggested formulation: check with a health worker before sharing.

- **Fund local fire crews** (historian, emergency manager) versus **much of the smoke comes from Laos** (skeptic). No source apportions Nan's smoke between local and cross-border fires.

WHAT NOBODY COVERED — BLIND SPOT

No lens could say how many bedridden and homebound people live in Nai Wiang, and the whole funding envelope scales with that number; the provincial long-term-care register has it. No lens quantified the smoke itself: the tracker counts heat only, and no study gives Nan's smoke deaths. And the cross-border share of Nan's smoke, which decides how much a local plan can do, is unmeasured.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are published at province level in the Climate Impact Lab tracker. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The location was taken as Nai Wiang subdistrict, the centre of Nan town (Nan Town Municipality), in Mueang Nan district. The tracker resolves it to Nan province (GADM THA.32_1). Mortality figures exist only at that scale; subdistrict detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Nan, 14 additional heat deaths June–September 2026 and 43 forecast October 2026 – March 2027 (October 3, November 4, December 1, January 2, February 4, March 29), 57 in total; Thailand, 15,000; the world, 857,400. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only; smoke is not included.

* Suggested formulation: check with a health worker before sharing.

- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Three verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial and governance actions were also checked for availability in Thailand, a named payer, a trigger threshold, and whether they already exist; food and water actions for published triggers and timing. The funding envelope was checked for real recipients, unit costs, and duplication, then re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Could not be consulted. The inventory page is client-side and returned no entries to the verifiers, and the download was not retrieved in this run. Actions tagged "not queryable" are unchecked against it, not shown to lack a precedent.
- **Money.** Costs are shown in Thai baht with US dollars in parentheses, at THB 33.701 = US$1 (Bank of Thailand weighted-average interbank rate, 30 October 2026).
- **Reader.** No role was given, so the For you section is written for a resident. The steps were taken from the verified actions and from what the panel found about residents here; any step not drawn from a ranked action was verified on the same three tests. The rest of the report is written for the district and its partners.
- **Health wording.** Passages where a wording error could hurt someone (cooling older people, sealed rooms in smoke, medicines, and heatstroke) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

* Suggested formulation: check with a health worker before sharing.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Costs marked "estimate" are the panel's planning figures, not quotes. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. The food lens found the El Niño food signal here weak and late (the WMO expects the event to last to February 2027), so no food action survived as a separate card. Several Thai sources returned errors (Daily News, Thairath, Matichon), so some figures rest on secondary reporting. No Nan-specific heat or smoke mortality estimate exists.

References · status after verification

Climate Impact Lab, El Niño Heat Mortality Tracker data v261026 (downloaded 30 Oct 2026) · confirmed

Filtered rooms: NHSO via MGR Online; Thairath, CMU room; The Standard; Suan Dok purifiers; Morishita 2018; Chiang Mai PM2.5 mortality; Matichon, 2,275 rooms · partially confirmed

Heat: JAMA 2025, fans in older adults; DDC heatstroke 2024; US heat alerts; volunteer heat role · partially confirmed

Governance and fire: Thai PBS, declaration criteria; Daily News, 2026 ban; forest closures; RECOFTC Nan; AP via Khaosod; Bangkok Post, spraying; Greenpeace 2023 · confirmed or corrected

Water and food: PRD, PWA Nan; Pattaya Mail 2012; FAO GIEWS; welfare card · corrected

Record: Al Jazeera 2006; Nation 2024; Nation 2025; Thaiger 2025; Nation 2026; Nan PHO dengue 2023 · confirmed or corrected

For you: N95 prices; Air4Thai stations; mask fit study · corrected

Exposure data · Climate Impact Lab, El Niño Heat Mortality Tracker (data v261026) · 451,000 at Risk (superseded) · Impact Map · Adaptation Inventory

Generated 30 October 2026 for Nai Wiang, Mueang Nan, Nan. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.